Type MS · Simulations and models: why we use them and their limits

Answer 1 of 4 · Question 4 part a · 2 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

Number			
4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Simulations and models: why we use them and their limits

Answer 1 of 4 · Question 4 part a · 2 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

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Answer 1 of 4 · Question 4 part a · 2 marks

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4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

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Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Simulations and models: why we use them and their limits

Answer 1 of 4 · Question 4 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · Simulations and models: why we use them and their limits

Answer 2 of 4 · Question 4 part a · 1 mark

2021 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Number		Guidance	
4(a)	Award **one** mark from: • Predict/model outcomes (1) • Investigate hypotheses (1) • Run experiments that cannot be run / are dangerous in real life (1) • Simulations can be speeded up/slowed down (1)		1

Type MS · Simulations and models: why we use them and their limits

Answer 2 of 4 · Question 4 part a · 1 mark

2021 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **two** marks from: • Simulations (may) use/collect large amounts of data (1) • Simulations (may) involve a large number of calculations (1) • Simulations may need more CPU cycles as the amount of data grows (1) • To be of value, the results must be calculated as quickly as possible (1) • Calculations may grow exponentially / become more complex as the simulation runs/is modified (1) • Simulation may output complex graphics (e.g. virtual world) (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)	<table><tr><th>NOT S</th><th>R AND W</th><th>(NOT S) OR (R AND W)</th></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • Correct values in NOT S column (1) • Correct values in R AND W column (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 1 and/or mark point 2	3

Type MS · Simulations and models: why we use them and their limits

Answer 2 of 4 · Question 4 part a · 1 mark

2021 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(d)(i)</td><td>Award two marks for a linked description such as:

• Instructions (and data) are stored in (main) memory (1)
• Instructions (and data) are fetched, decoded, and executed (in a sequence by the CPU) (1)</td><td>Must include storage and execution for 2 marks</td><td>2</td></tr>
<tr><td>4(d)(ii)</td><td>Award two marks from:

• Program counter / PC (1)
• Current instruction (register) / CIR (1)
• Memory address (register) / MAR (1)
• Memory buffer/data (register) /MBR/MDR (1)
• Accumulator/ACC (1)</td><td></td><td>2</td></tr>
<tr><td>4(d)(iii)</td><td>Award two marks for a linked description such as:

Address of instruction carried on address bus (to main memory) (1) contents at that address carried back (to the CPU) on the data bus (1)</td><td></td><td>2</td></tr>
<tr><td>4(d)(iv)</td><td>Award two marks from:

<table><tr><th>Change made</th><th>Effect</th></tr><tr><td>Increasing the width of the data bus</td><td>Increases the number of bits/data/ size of word that can be transferred at one time / increases rate of data transfer (1)</td></tr><tr><td>Increasing the width of the address bus</td><td>Increases the number of memory addresses / increases the maximum amount of memory that can be addressed (1)</td></tr></table></td><td>Do not accept increases bandwidth without further explanation</td><td>2</td></tr>
</table>

Type MS · Simulations and models: why we use them and their limits

Answer 2 of 4 · Question 4 part a · 1 mark

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Question Number	Answer	Additional Guidance	Mark

Type MS · Simulations and models: why we use them and their limits

Answer 3 of 4 · Question 4 part b · 2 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

Number		Guidance	
4(a)	Award **one** mark from: • Predict/model outcomes (1) • Investigate hypotheses (1) • Run experiments that cannot be run / are dangerous in real life (1) • Simulations can be speeded up/slowed down (1)		1

Type MS · Simulations and models: why we use them and their limits

Answer 3 of 4 · Question 4 part b · 2 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
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Question Number	Answer	Additional Guidance	Mark
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Type MS · Simulations and models: why we use them and their limits

Answer 3 of 4 · Question 4 part b · 2 marks

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Answer 3 of 4 · Question 4 part b · 2 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · Simulations and models: why we use them and their limits

Answer 4 of 4 · Question 3 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Simulations and models: why we use them and their limits

Answer 4 of 4 · Question 3 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

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Answer 4 of 4 · Question 3 part a · 2 marks

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Question number	Answer	Additional guidance	Mark
3(d)	Award **one** mark for each of: • All file blocks of each file are in numeric sequence/one after another in numeric order (1) • All blank space is together at the front/at the end (1) Examples: W1 W2 W3 W4 Z1 Z2 Z3 Z1 Z2 Z3 W1 W2 W3 W4 Z1 Z2 Z3 W1 W2 W3 W4 W1 W2 W3 W4 Z1 Z2 Z3	Note: Order of letter and number does not matter. W1 or 1W are equally acceptable	2

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Answer 4 of 4 · Question 3 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

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Question Number	Answer	Additional Guidance	Mark